Prime values and consecutive-prime clusters of $n^5 - (n-1)^5$: Computation, heuristics, and large probable prime generation

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Abstract

We report on an exhaustive computational search for prime values of the quintic difference polynomial $Q(n) = n^5 - (n-1)^5$ over the range $2 \le n \le 10^8$, yielding 5,179,467 primes. We derive the Bateman–Horn constant $C_Q = 3.678113109 \pm 3 \times 10^{-9}$ from the factorization structure of Q over the cyclotomic field $\mathbb{Q}(\zeta_5)$, and verify that the observed prime counts agree with the prediction $\pi_Q(N) \sim C_Q \int_2^N dt/\log Q(t)$ to within 0.01% at $N = 5 \times 10^7$. We analyze the distribution of consecutive-prime k-tuplets (sequences $n, n{+}1, \ldots, n{+}k{-}1$ for which $Q(n), Q(n{+}1), \ldots, Q(n{+}k{-}1)$ are all prime) and prove that 7-tuplets cannot exist, while 6-tuplets can and do occur. Extended searches up to $n \approx 1.9 \times 10^9$ have located 25 sextuplets, 458 quintuplets, and 11,063 quadruplets. Hardy–Littlewood-type singular series for these k-tuplets are computed and shown to match observed counts. Finally, we describe a three-stage pipeline (C++ modular sieve, target construction, PFGW probable-prime testing) that has produced 175 probable primes $Q(n)$ with $\sim$10,000 digits, two with $\sim$50,000 digits, and one with $\sim$100,000 digits. One of the 10,000-digit candidates has been rigorously certified prime via ECPP, yielding a 9,997-digit proven prime of the form $n^5 - (n-1)^5$.

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1 Introduction

For a fixed polynomial $f \in \mathbb{Z}[x]$, the question of how often $f(n)$ takes prime values as n ranges over the positive integers is classical and largely open. Aside from the linear case $f(n) = an + b$ (Dirichlet's theorem on primes in arithmetic progressions), no polynomial of degree ≥ 2 has been proved to represent infinitely many primes. Nevertheless, the Bateman–Horn conjecture [2] provides a precise quantitative heuristic, and extensive numerical evidence supports it for many families; see Sorenson [16] for a recent survey.

In this paper we study the polynomial

$$Q(n) \;=\; n^5 - (n-1)^5 \;=\; 5n^4 - 10n^3 + 10n^2 - 5n + 1, \tag{1}$$

which arises naturally as the fifth-power analogue of the well-known case $n^2 - (n-1)^2 = 2n - 1$, whose prime values are simply the odd primes. We note the symmetry $Q(n) = Q(1-n)$, which follows from the substitution $n \mapsto 1 - n$ in (1); consequently, the study of prime values for negative n reduces trivially to the positive case. The cubic case $n^3 - (n-1)^3 = 3n^2 - 3n + 1$ has been investigated as a source of large primes by Dubner and others [5], but the quintic case appears not to have received systematic attention.

Two features make $Q(n)$ particularly attractive:

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(i) **Arithmetic rigidity.** The factorization of Q is governed by the fifth cyclotomic polynomial Φ_5, which forces every prime divisor of $Q(n)$ to be congruent to 1 (mod 5) (see Corollary 2). This eliminates three-quarters of all primes from the sieve, producing a large Bateman–Horn constant $C_Q \approx 3.678$ — meaning $Q(n)$ is roughly 3.68 times as likely to be prime as a "random" integer of the same size.

(ii) **Tuplet clustering.** The same arithmetic rigidity that boosts overall prime density also permits long runs of consecutive n-values for which $Q(n)$ is simultaneously prime. We call a maximal run of k consecutive integers $\{n, n+1, \ldots, n+k-1\}$ all yielding primes a *k-tuplet*. We prove that $k \leq 6$ is a hard upper bound imposed by the roots of Q modulo 11, and exhibit data showing that this bound is achieved.

The paper is organized as follows. Section 2 develops the arithmetic of $Q(n)$ modulo primes, establishing the root structure and admissibility constraints. Section 3 derives the Bateman–Horn prediction and compares it with our computational data. Section 4 analyzes the prime gap distribution. Section 5 treats k-tuplets and their Hardy–Littlewood heuristics. Section 6 describes the algorithmic pipeline for generating large probable primes of the form $Q(n)$, and Section 7 concludes with open problems.

2 Arithmetic of $Q(n)$ modulo primes

2.1 Cyclotomic structure

The identity

$$n^5 - (n-1)^5 \;=\; (n-(n-1)) \cdot \sum_{j=0}^{4} n^j (n-1)^{4-j} \;=\; \sum_{j=0}^{4} n^j (n-1)^{4-j} \tag{2}$$

shows that $Q(n)$ is the norm form arising from the factorization of $x^5 - 1$. More precisely, setting $x = n/(n-1)$, we have

$$Q(n) \;=\; (n-1)^4\, \Phi_5\!\left(\frac{n}{n-1}\right), \tag{3}$$

where $\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$ is the fifth cyclotomic polynomial. The splitting behavior of Φ_5 modulo a prime p is determined by the order of p in $(\mathbb{Z}/5\mathbb{Z})^*$:

Proposition 1. *Let p be a prime and let $\omega_Q(p)$ denote the number of solutions $n \in \{0, 1, \ldots, p-1\}$ to $Q(n) \equiv 0 \pmod{p}$. Then:*

(a) *If $p = 5$, then $\omega_Q(5) = 0$. In fact, $Q(n) \equiv 1 \pmod 5$ for all n.*

(b) *If $p \equiv 1 \pmod 5$, then $\omega_Q(p) = 4$.*

(c) *If $p \not\equiv 0, 1 \pmod 5$, then $\omega_Q(p) = 0$.*

Proof. (a) By Fermat's little theorem, $n^5 \equiv n \pmod 5$ for all n, so $Q(n) = n^5 - (n-1)^5 \equiv n - (n-1) = 1 \pmod 5$.

(b)–(c) By (3), for $p \nmid n-1$ we have $Q(n) \equiv 0 \pmod p$ if and only if $\Phi_5(n/(n-1)) \equiv 0 \pmod p$, i.e., $n/(n-1)$ is a primitive fifth root of unity modulo p. When $p \equiv 1 \pmod 5$, the group $(\mathbb{Z}/p\mathbb{Z})^*$ contains exactly four primitive fifth roots of unity; each yields a unique $n \equiv \zeta/(\zeta - 1) \pmod p$ with $\zeta \neq 1$. When $p \not\equiv 1 \pmod 5$ and $p \neq 5$, there are no primitive fifth roots of unity in $(\mathbb{Z}/p\mathbb{Z})^*$, so Φ_5 has no roots. One verifies separately that $n \equiv 1 \pmod p$ does not yield $Q(n) \equiv 0 \pmod p$ (since $Q(1) = 1$). □

Corollary 2. *Every prime factor of $Q(n)$ is congruent to 1 (mod 5). In particular, $Q(n)$ is never divisible by $2, 3, 7$, or any prime $p \not\equiv 1 \pmod 5$.*

Proof. If $p \mid (n-1)$ then $Q(n) \equiv 1 \pmod{p}$, so $p \nmid Q(n)$. For $p \nmid (n-1)$, Proposition 1 gives the result. $\square$

The smallest prime $\equiv 1 \pmod{5}$ is $p = 11$. Its four roots are:

$$Q(n) \equiv 0 \pmod{11} \iff n \equiv 4, 5, 7, 8 \pmod{11}. \tag{4}$$

Thus $Q(n)$ can be prime only when $n \bmod 11 \in \{0, 1, 2, 3, 6, 9, 10\}$, a set of 7 residues.

Table 1 lists the roots for the first several primes $p \equiv 1 \pmod{5}$.

Table 1: Roots of $Q(n) \equiv 0 \pmod{p}$ for primes $p \equiv 1 \pmod{5}$

p	Roots $\{r : Q(r) \equiv 0\}$	Largest consecutive safe run	# safe residues
11	$\{4, 5, 7, 8\}$	6 $(9, 10, 0, 1, 2, 3)$	7
31	$\{2, 10, 22, 30\}$	11	27
41	$\{9, 12, 30, 33\}$	17	37
61	$\{16, 24, 38, 46\}$	30	57
71	$\{4, 19, 53, 68\}$	33	67
101	$\{27, 29, 73, 75\}$	52	97

Note: For every prime $p \equiv 1 \pmod{5}$, the number of safe residues is exactly $p - 4$.

2.2 Admissibility and the maximal tuplet length

Theorem 3. *There is no integer n such that $Q(n), Q(n+1), \ldots, Q(n+6)$ are all prime. That is, 7-tuplets for Q do not exist. Moreover, 6 is the maximal tuplet length, and this bound is achieved.*

Proof. By (4), the set of "forbidden" residues modulo 11 is $F = \{4, 5, 7, 8\}$. The complementary "safe" residues are $S = \{0, 1, 2, 3, 6, 9, 10\}$. The maximal length of a run of consecutive integers drawn entirely from S is determined by the largest gap in F when the residues $\{0, \ldots, 10\}$ are arranged cyclically. The gaps between consecutive elements of F are $5 - 4 = 1$, $7 - 5 = 2$, $8 - 7 = 1$, and $(4 + 11) - 8 = 7$. The largest gap is 7, corresponding to the run $9, 10, 0, 1, 2, 3$ of length 6. Hence any 7 consecutive integers must contain at least one element of F, making Q divisible by 11 at that point.

Conversely, computations described in Section 5 exhibit 25 sextuplets in the range $n \leq 1.9 \times 10^9$, the smallest at $n = 10{,}291{,}136$. $\square$

3 The Bateman–Horn prediction

3.1 Statement and constant computation

The Bateman–Horn conjecture [2] predicts that for an irreducible polynomial $f \in \mathbb{Z}[x]$ with positive leading coefficient and no fixed prime divisor,

$$\pi_f(N) \;:=\; \#\{n \leq N : f(n) \text{ is prime}\} \;\sim\; C_f \int_2^N \frac{dt}{\log f(t)}, \tag{5}$$

where the product

$$C_f \;=\; \prod_{p \text{ prime}} \frac{1 - \omega_f(p)/p}{1 - 1/p} \tag{6}$$

converges to a positive constant (here $\omega_f(p)$ denotes the number of roots of f modulo p).

For $f = Q$, Proposition 1 gives:

$$C_Q \;=\; \frac{5}{4} \cdot \prod_{\substack{p\neq 5\\ p\not\equiv 1\,(5)}} \frac{p}{p-1} \cdot \prod_{p\equiv 1\,(5)} \frac{p-4}{p-1}. \tag{7}$$

The convergence follows from Dirichlet's theorem on primes in arithmetic progressions: primes $p \equiv 1 \pmod 5$ have natural density $1/4$ among all primes, so the average of $\omega_Q(p)$ is $4\cdot\frac{1}{4}+0\cdot\frac{3}{4} = 1$, and the logarithmic contributions

$$\sum_p \frac{\omega_Q(p)-1}{p}$$

converge by partial summation and the Siegel–Walfisz theorem (see [8, Ch. 17] for the general theory).

We evaluate (7) numerically by computing the partial Euler product $C_Q(P) = \prod_{p\le P}(1 - \omega_Q(p)/p)/(1-1/p)$ over all primes $p \le P$. The tail error admits a precise bound: writing $f(p) = \log[(1-\omega_Q(p)/p)/(1-1/p)]$ and using $\log(1-x) = -x - x^2/2 + O(x^3)$, we obtain

$$f(p) = \frac{1-\omega_Q(p)}{p} + \frac{1-\big(\omega_Q(p)\big)^2}{2p^2} + O(p^{-3}). \tag{8}$$

The first-order terms $\sum_{p>P}(1-\omega_Q(p))/p$ cancel by Dirichlet's theorem, since ω_Q averages to 1 over primes in each residue class modulo 5. The dominant tail contribution comes from the second-order terms: for $p \equiv 1 \pmod 5$ (where $\omega_Q = 4$), the coefficient is $(1-16)/(2p^2) = -\frac{15}{2p^2}$, while for $p \not\equiv 1 \pmod 5$ (where $\omega_Q = 0$), it is $1/(2p^2)$. By the prime-number theorem for arithmetic progressions (each class carries density $1/4$),

$$\log C_Q - \log C_Q(P) = \Big(-\tfrac{15}{8}+\tfrac{3}{8}\Big)\sum_{p>P}\frac{1}{p^2} + O\Big(\sum_{p>P}\frac{1}{p^3}\Big) = -\frac{3}{2}\sum_{p>P}\frac{1}{p^2} + O(P^{-2}\log^{-1}P).$$

By partial summation with $\pi(t) \sim t/\log t$, one has $\sum_{p>P} 1/p^2 \sim 1/(P\log P)$, so

$$|\log C_Q - \log C_Q(P)| \;\sim\; \frac{3}{2P\log P}.$$

Extending the computation to all primes $p \le 10^8$ yields (Table 2)

$$C_Q \;=\; 3.678113109 \pm 3\times 10^{-9}, \tag{9}$$

where the error bound follows from the tail estimate above with $P = 10^8$ and $\log P \approx 18.4$. Higher precision could be obtained by expressing C_Q in closed form via Dirichlet L-functions $L(1,\chi)$ for the nontrivial characters χ mod 5 (cf. [3, 9]). The relevant L-values are:

$$L(1,\chi_2) = \frac{2\log\varphi}{\sqrt{5}} = 0.43040894096\ldots \quad \text{(class number formula, } \varphi = \tfrac{1+\sqrt{5}}{2}\text{)},$$
$$|L(1,\chi_1)|^2 = \frac{[\psi(\frac{1}{5})-\psi(\frac{4}{5})]^2 + [\psi(\frac{2}{5})-\psi(\frac{3}{5})]^2}{25} = 0.78956835209\ldots,$$

where ψ is the digamma function and χ_1 is the primitive character of order 4 modulo 5. Full evaluation of C_Q to arbitrary precision via these values (combined with an accelerated Euler product over small primes) is straightforward in a computer algebra system such as PARI/GP, and would yield a value consistent with (9).

Table 2: Convergence of the partial Euler product for C_Q

Primes $p \leq$	Partial product
10^2	3.5557
10^3	3.6746
10^4	3.6715
10^5	3.6777
10^6	3.6777
10^7	3.6781
10^8	3.678113109

3.2 Comparison with data

We computed all $n \leq 10^8$ for which $Q(n)$ is prime by a segmented sieve exploiting the root structure of Proposition 1: for each prime $p \equiv 1 \pmod 5$ up to a suitable bound, we mark the four residue classes where $Q(n) \equiv 0 \pmod p$, then test survivors for primality (see [6] for related sieve methodology). The search found $\pi_Q(10^8) = 5{,}179{,}467$ primes. Table 3 compares the observed counts with the Bateman–Horn prediction $C_Q \int_2^N dt/\log Q(t)$ using $C_Q = 3.6781$.

Table 3: Comparison of $\pi_Q(N)$ with Bateman–Horn prediction

N	$\pi_Q(N)$	$C_Q \int_2^N \frac{dt}{\log Q(t)}$	Ratio
10^3	149	151.1	0.9864
10^4	1,101	1,088.6	1.0114
10^5	8,489	8,518.3	0.9966
10^6	69,977	70,061.2	0.9988
2×10^6	133,203	133,036.3	1.0013
5×10^6	312,522	311,772.9	1.0024
10^7	596,022	595,382.1	1.0011
2×10^7	1,139,867	1,139,339.1	1.0005
5×10^7	2,694,860	2,694,614.7	1.0001

The agreement is striking: at $N = 5 \times 10^7$, the observed count exceeds the prediction by only 245 primes out of 2.69 million, a relative error of 0.009%.

3.3 Residue class uniformity

As a further consistency check, Corollary 2 predicts that prime-generating values of n should be uniformly distributed among the 7 safe residue classes modulo 11. Table 4 confirms this prediction to high precision.

4 Prime gap distribution

Let $n_1 < n_2 < \cdots$ be the sequence of positive integers for which $Q(n_i)$ is prime, and set $g_i = n_{i+1} - n_i$. Over 5,179,466 gaps in our dataset, we find a mean gap of $\bar{g} = 19.31$ with a maximum of 284. (For comparison with gap statistics for ordinary primes, see Oliveira e Silva et al. [10].)

Table 4: Distribution of $n \pmod{11}$ among all 5,179,467 primes $Q(n)$

n mod 11	Count	Status
0	738,642	safe
1	739,726	safe
2	740,210	safe
3	739,507	safe
4	0	root
5	0	root
6	740,038	safe
7	0	root
8	0	root
9	740,500	safe
10	740,844	safe

Expected per safe class: 739,924. Maximum deviation: 0.12%.

4.1 Raw gap frequencies

The gap distribution displays a pronounced mod-11 structure inherited from the forbidden residues (4). Table 5 shows frequencies for small gaps.

Table 5: Gap frequencies $g_i = n_{i+1} - n_i$ for small values

Gap	Count	Frequency	Gap	Count	Frequency
1	288,450	5.57%	7	182,678	3.53%
2	218,017	4.21%	8	237,959	4.59%
3	290,431	5.61%	9	165,254	3.19%
4	196,226	3.79%	10	190,263	3.67%
5	139,104	2.69%	11	266,046	5.14%
6	141,793	2.74%	22	144,659	2.79%

Several features are noteworthy. Gaps $g = 1$ and $g = 3$ are the most common, reflecting the pair structure of safe residues modulo 11: the safe set $\{0, 1, 2, 3, 6, 9, 10\}$ contains the consecutive blocks $\{0, 1, 2, 3\}$ and $\{9, 10\}$, favoring short gaps. The prominence of $g = 11$ (and multiples thereof) arises because stepping by 11 preserves the residue class modulo 11.

4.2 Normalized gaps and statistical testing

To remove the slow variation in gap size, we normalize: $\hat{g}_i = g_i/E_i$ where $E_i = (4 \log n_i)/C_Q$ is the local expected gap from the Bateman–Horn density. The mean of $\hat{g}$ is 1.023 (close to the theoretical 1), with standard deviation 0.974.

Under the simplest heuristic (Cramér model), consecutive primes of a polynomial behave like independent events with local probability $1/E_i$, predicting that $\hat{g}$ follows a standard exponential distribution with density e^{-x}. Table 6 compares the empirical distribution with this prediction and with a *mod*-11 *corrected* model that accounts for the discrete constraint imposed by the forbidden residues $\{4, 5, 7, 8\} \pmod{11}$.

The corrected model is defined as follows. For a raw gap g starting at a safe residue r mod 11, the transition weight $w(g) = \frac{1}{7}\#\{r \in S : (r + g) \bmod 11 \in S\}$ captures the fraction of safe-to-safe transitions, where $S = \{0, 1, 2, 3, 6, 9, 10\}$. Gaps $g \equiv 0 \pmod{11}$ receive weight $w = 1$, while $g \equiv 5, 6 \pmod{11}$ receive the minimum $w = 3/7$. The corrected bin probabilities are

Table 6: Empirical distribution of normalized gaps $\hat{g}_i = g_i/E_i$, where $E_i = (4\log n_i)/C_Q$ is the local expected gap

Bin	Observed	Fraction	Exponential	Mod-11 corrected
[0, 0.25)	988,981	0.191	0.221	0.192
[0.25, 0.50)	835,415	0.161	0.172	0.158
[0.50, 0.75)	860,533	0.166	0.134	0.170
[0.75, 1.00)	485,622	0.094	0.104	0.095
[1.00, 1.50)	821,858	0.159	0.145	0.143
[1.50, 2.00)	481,813	0.093	0.088	0.100
[2.00, 3.00)	457,348	0.088	0.086	0.089
[3.00, 4.00)	158,962	0.031	0.031	0.034
[4.00, 6.00)	77,769	0.015	0.016	0.016
≥ 6.00	10,140	0.002	0.002	0.003

$P_{\mathrm{corr}}([a,b)) = Z^{-1}\sum_{g=\lceil aE\rceil}^{\lfloor bE\rfloor} w(g)\, e^{-g/E}/E$ with $E = \bar{g} = 19.31$ and Z a normalization constant.

With 5.18×10^6 data points, classical χ^2 tests are uninformative: even tiny model deviations produce enormous test statistics that always reject. We therefore quantify the fit using sample-size-independent effect-size metrics. The *total variation distance* (TVD) and *Kullback–Leibler divergence* from the observed distribution to each model are:

$$\begin{aligned} \mathrm{TVD}_{\mathrm{exp}} &= 0.054, \qquad & D_{\mathrm{KL}}^{\mathrm{exp}} &= 7.6\times 10^{-3}, \\ \mathrm{TVD}_{\text{mod-11}} &= 0.019, \qquad & D_{\mathrm{KL}}^{\text{mod-11}} &= 1.6\times 10^{-3}. \end{aligned} \tag{10}$$

The mod-11 correction reduces the TVD by a factor of 2.8 and the KL divergence by a factor of 4.8, confirming that $p = 11$ is the dominant source of deviation from exponential behavior. The residual TVD of 0.019 is attributable to the higher sieving primes $p = 31, 41, 61, \ldots$, each of which imposes additional transition-weight modulations on the gap distribution; a full multi-prime model incorporating all primes $p \equiv 1 \pmod 5$ up to P would, in the limit $P \to \infty$, converge to the exact gap law.

5 Consecutive-prime tuplets

5.1 Definitions and counts

Definition 4. A *k-tuplet* for Q is a maximal block of k consecutive integers $\{n, n{+}1, \ldots, n{+}k{-}1\}$ such that $Q(n), Q(n{+}1), \ldots, Q(n{+}k{-}1)$ are all prime. When counting non-maximally (allowing overlaps from longer runs), we write $T_k(N)$ for the number of starting positions $n \leq N$.

Table 7 summarizes our findings. The data for $n \leq 10^8$ come from the exhaustive search described above; the data for $n \leq 1.9\times 10^9$ come from an extended sieve-based search with PFGW verification of primality for candidates near known quadruplets.

5.2 Hardy–Littlewood heuristic for k-tuplets

To predict $T_k(N)$, we generalize the Bateman–Horn conjecture to the k-tuple of polynomials $f_j(n) = Q(n+j)$ for $j = 0, 1, \ldots, k-1$. Following the framework of Hardy and Littlewood [7], the singular series becomes

$$\mathfrak{S}_k \;=\; \prod_p \frac{1-\omega_k(p)/p}{(1-1/p)^k}, \tag{11}$$

Table 7: k-tuplet counts (maximal runs)

k	$n \le 10^8$	$n \le 1.9 \times 10^9$	Smallest n
2 (pairs)	254,833	—	2
3 (triplets)	15,307	—	427
4 (quadruplets)	924	11,063	383
5 (quintuplets)	54	458	1,915,329
6 (sextuplets)	3	25	10,291,136
≥ 7	0	0	(impossible)

where $\omega_k(p) = \#\{n \bmod p : Q(n+j) \equiv 0 \pmod p \text{ for some } 0 \le j < k\}$ is the number of "unsafe" starting positions modulo p.

For primes $p \not\equiv 1 \pmod 5$ (and $p \ne 5$), we have $\omega_k(p) = 0$, so the factor is $(1 - 1/p)^{-k}$. For $p \equiv 1 \pmod 5$, the four roots of $Q \pmod p$ contribute: $\omega_k(p) = p - s_k(p)$, where $s_k(p)$ is the number of starting positions modulo p for which the window $\{n, \ldots, n+k-1\}$ avoids all four roots.

The predicted count is

$$T_k(N) \sim \mathfrak{S}_k \int_2^N \frac{dt}{(\log Q(t))^k} \,. \tag{12}$$

Remark 5. More precisely, the Bateman–Horn conjecture applied to the k-tuple $(Q(n), Q(n+1), \ldots, Q(n+k-1))$ gives

$$T_k(N) \sim \mathfrak{S}_k \int_2^N \frac{dt}{\log Q(t) \cdot \log Q(t+1) \cdots \log Q(t+k-1)} \,.$$

Since $Q(t) = 5t^4(1 + O(1/t))$, we have $\log Q(t+j) = \log Q(t) + O(k/t)$ for fixed k, and the two integral forms agree asymptotically. At $N = 10^8$, the relative difference between $\prod_{j=0}^{k-1} \log Q(t+j)$ and $(\log Q(t))^k$ is of order $k^2/(t \log t) < 10^{-5}$ for all $t \ge 100$, so the simpler form (12) introduces negligible error for the data in Table 8. For smaller N or larger k, the product form would be preferable.

Table 8 compares these predictions with the observed *overlapping* counts $T_k(10^8)$. Note that these differ from the maximal-run counts in Table 7: for example, a single maximal 6-tuplet at n contributes one entry to the "$k = 6$" row of Table 7, but contributes to T_6, T_5, T_4, T_3, and T_2 in Table 8 (providing respectively 1, 2, 3, 4, and 5 overlapping sub-tuplets).

Table 8: Singular series and tuplet predictions at $N = 10^8$ (overlapping counts T_k)

k	$\mathfrak{S}_k$	$\int_2^{10^8} \frac{dt}{(\log Q)^k}$	Predicted	Observed
2	14.42	19,903	286,951	288,450
3	60.81	282.9	17,200	17,329
4	255.0	4.050	1,033	1,041
5	903.2	0.0594	53.6	60
6	2,220	0.0011	2.4	3

For $k = 2, 3, 4$, the predictions agree with the data to within 1%, providing strong evidence for the Hardy–Littlewood model. The $k = 5, 6$ cases show larger deviations (12% and 26% respectively), as expected from the small sample sizes.

Remark 6. The vanishing of $\mathfrak{S}_7$ provides an independent, heuristic-free confirmation of Theorem 3: at $p = 11$, we have $s_7(11) = 0$, so $\omega_7(11) = 11$, and the factor $(1 - 11/11)/(1 - 1/11)^7 = 0$ forces $\mathfrak{S}_7 = 0$.

5.3 Distribution of sextuplets

The 25 sextuplets found up to $n \approx 1.9 \times 10^9$ begin at the following values of n:

$$\begin{gathered}
10{,}291{,}136,\ \ 45{,}197{,}062,\ \ 89{,}166{,}471,\ \ 139{,}663{,}665,\ \ 144{,}073{,}631,\ \ 150{,}737{,}728,\\
159{,}933{,}002,\ \ 169{,}996{,}638,\ \ 181{,}637{,}322,\ \ 226{,}237{,}427,\ \ 233{,}892{,}349,\ \ 308{,}374{,}240,\\
506{,}989{,}173,\ \ 548{,}169{,}675,\ \ 570{,}232{,}177,\ \ 608{,}369{,}804,\ \ 843{,}734{,}362,\ \ 865{,}999{,}594,\\
1{,}055{,}031{,}536,\ \ 1{,}123{,}085{,}445,\ \ 1{,}188{,}401{,}113,\ \ 1{,}393{,}470{,}692,\ \ 1{,}625{,}095{,}965,\\
1{,}686{,}424{,}793,\ \ 1{,}932{,}852{,}337.
\end{gathered}$$

A sextuplet at n requires $n, n{+}1, \ldots, n{+}5$ to all be safe modulo 11. Since the unique maximal block of 6 consecutive safe residues is $\{9, 10, 0, 1, 2, 3\}$, every sextuplet must satisfy $n \equiv 9 \pmod{11}$. We have verified that all 25 values listed above do satisfy this necessary condition.

6 Large probable prime generation

6.1 Three-stage pipeline

The arithmetic rigidity of $Q(n)$ (Corollary 2) makes it an excellent candidate for generating titanic probable primes ($\geq$ 1,000 digits) and larger. We developed a three-stage pipeline:

Stage 1: Modular sieve (C++). For a chosen base n_0 (e.g., $n_0 = 10^{2499}$ to target 10,000-digit values of Q), we sieve the interval $\{n_0, n_0{+}1, \ldots, n_0{+}R{-}1\}$ for offsets i such that $Q(n_0{+}i)$ has no prime factor $p \leq B$ with $p \equiv 1 \pmod 5$.

For each such prime p, we compute the four roots $r_1, \ldots, r_4$ of $Q \pmod p$ using a primitive fifth root of unity, then mark all i with $n_0 + i \equiv r_j \pmod p$. With $B = 10^9$ and $R = 1{,}200{,}000$, this eliminates approximately 99.97% of candidates, leaving several thousand "survivors."

The sieve is implemented in C++ using 128-bit arithmetic for modular reductions (cf. [14, 15] for background on prime-number sieves) and runs in approximately one hour on a single core (AMD Ryzen 9 5950X, 3.4 GHz).

Stage 2: Target construction (Python). The surviving offsets are expanded into full decimal representations of $Q(n_0 + i)$, producing numbers of $\sim 4d$ digits where d is the number of digits in n_0. These are formatted for input to PFGW.

Stage 3: PRP testing (PFGW). Each target is subjected to a strong Fermat probable-prime test using PFGW [11], with base 3 and a subsequent Brillhart–Lehmer–Selfridge (BLS) partial factorization test when applicable. A 10,000-digit candidate requires approximately 30 seconds; a 100,000-digit candidate requires several hours.

Remark 7. The numbers produced by the pipeline above are initially *probable primes* (PRPs) based on a strong Fermat test. To obtain a rigorous primality proof, we subjected the smallest candidate $Q(10^{2499}{+}3{,}469)$ to the ECPP algorithm [1] using the implementation Primo 4.3.3 [13]. The computation required 286,269 seconds of wall-clock time ($\approx$ 3.3 days; 43.4 CPU-days across multiple cores), producing a certificate chain of 887 steps that unconditionally proves $Q(10^{2499} + 3{,}469)$ is prime. This 9,997-digit number is the largest proven prime of the form $n^5 - (n-1)^5$ currently listed in the Prime Pages database [12]. The Primo certificate file is included in the supplementary materials. The remaining 174 candidates in Appendix A, as well as the 50,000- and 100,000-digit PRPs, await certification.

6.2 Results

Table 9 summarizes the probable primes found to date.

In the 10,000-digit search, the sieve covered offsets $0 \leq i < 1{,}200{,}000$ from the base $n_0 = 10^{2499}$, with sieve depth $B = 10^9$. Subsequent PFGW testing of all survivors identified 175

Table 9: Probable primes (PRPs) of the form $Q(n) = n^5 - (n-1)^5$

Digit class	Base n_0	Offset range	PRPs	Status
$\sim$ 10,000	10^{2499}	$3{,}469 \le k \le 1{,}162{,}743$	175	$k = 3{,}469$ certified prime
$\sim$ 50,000	10^{12499}	$k \in \{5{,}730,\ 29{,}658\}$	2	PRP
$\sim$ 100,000	10^{24999}	$k = 25{,}098$	1	PRP

probable primes $Q(10^{2499} + k)$, each having 9,997 decimal digits. The complete list of all 175 offsets is provided in Appendix A. As described in Remark 7, the smallest of these—the 9,997-digit number

$$Q(10^{2499} + 3{,}469) \;=\; (10^{2499} + 3{,}469)^5 - (10^{2499} + 3{,}468)^5$$

—has been rigorously certified prime via ECPP.

The mean spacing between consecutive PRPs is $\approx 6{,}662$, consistent with the Bateman–Horn density $C_Q/\log Q(n_0) \approx 1.6 \times 10^{-4}$. We note the close pair $k = 106{,}459$ and $k = 106{,}520$ (gap 61), suggesting that tuplet-like clustering persists even at the 10,000-digit scale.

For the 50,000-digit class, two probable primes emerged from a similar pipeline. The 100,000-digit probable prime

$$Q(10^{24999} + 25{,}098) \;=\; (10^{24999} + 25{,}098)^5 - (10^{24999} + 25{,}097)^5$$

has 99,997 decimal digits and is the largest known (probable) prime of the form $n^5 - (n-1)^5$ as of March 2026 [12].

7 Concluding remarks

The polynomial $Q(n) = n^5 - (n-1)^5$ displays a remarkably clean arithmetic structure rooted in the fifth cyclotomic field. The restriction that all prime divisors satisfy $p \equiv 1 \pmod 5$ boosts the prime density by a factor of $C_Q \approx 3.678$, enables the existence of consecutive-prime sextuplets, and simultaneously imposes the sharp bound $k \le 6$ on tuplet length. All quantitative predictions from the Bateman–Horn and Hardy–Littlewood frameworks are confirmed by the data to high precision.

Several natural directions for further work include:

(i) *Further primality certifications.* Having certified $Q(10^{2499} + 3{,}469)$ as a 9,997-digit proven prime (Remark 7), we plan to certify additional candidates from Appendix A. Each ECPP run at this scale requires ≈ 3 days of wall-clock time (Primo 4.3.3 on a multi-core workstation), making batch certification of all 175 candidates a significant but tractable computational project.

(ii) *Megaprime search.* The sieve pipeline scales well: replacing the base $n_0 = 10^{2499}$ with $n_0 = 10^{249999}$ would target 10^6-digit probable primes. The sieve cost is dominated by the number of active primes $p \le B$ with $p \equiv 1 \pmod 5$, which is independent of n_0; only the PRP testing cost grows (roughly as d^2 for d-digit numbers under FFT-based arithmetic [4]).

(iii) *Higher cyclotomic analogues.* For odd k, the polynomial $n^k - (n-1)^k$ is governed by Φ_k, and the sieving restriction is $p \equiv 1 \pmod k$. The cases $k = 7, 11, 13$ would give even stronger density boosts (C_f grows with k) but require deeper sieving and present more complex root structures.

(iv) *Tuplet asymptotics.* The singular series $\mathfrak{S}_k$ for Q grow rapidly with k (Table 8), but the integral factor $\int dt/(\log Q)^k$ shrinks faster. A careful analysis of when $\mathfrak{S}_k \int dt/(\log Q)^k$ drops below 1 (expected number < 1) would give an effective "last expected k-tuplet" bound.

(v) *CRT-based targeted search.* Rather than sieving a contiguous interval, one could use the Chinese Remainder Theorem to solve a system of congruences $n \not\equiv r_j \pmod{p}$ simultaneously for many primes p, directly computing candidate n-values that survive deep sieving. This approach, if combined with lattice reduction, could accelerate the search for tuplets or large primes by orders of magnitude. We leave this as an open problem.

A Complete list of 10,000-digit PRP offsets

The following is the complete list of all 175 offsets k ($3{,}469 \le k \le 1{,}162{,}743$) for which $Q(10^{2499}+k)$ is a probable prime, as described in Section 6:

3469, 6073, 14791, 15367, 20398, 26943, 44759, 48337, 49436, 50368, 60716, 61899, 64870, 65769, 71174, 73025, 87603, 88183, 91080, 93129, 94912, 100905, 105403, 106459, 106520, 109353, 115504, 117292, 131871, 135981, 137534, 138028, 139571, 142308, 155852, 156578, 162669, 179711, 182412, 185742, 187007, 190434, 203823, 205594, 207126, 208795, 214215, 216989, 223717, 229791, 230914, 235979, 238029, 242233, 242396, 243111, 245819, 261320, 262346, 263803, 266465, 281890, 283375, 289677, 293729, 300741, 301347, 307443, 311049, 325149, 328230, 328680, 333465, 334550, 335280, 351457, 364478, 384056, 384285, 387584, 392416, 393658, 411127, 418036, 419267, 426224, 429397, 437129, 444198, 445412, 445939, 483770, 495067, 496221, 532085, 541966, 558371, 559053, 568659, 575761, 578951, 587286, 595302, 598946, 604496, 615156, 620279, 621807, 623113, 644365, 646958, 650606, 650928, 676413, 677717, 681672, 684941, 696148, 709895, 722259, 745011, 747585, 747809, 758768, 788690, 789870, 796358, 796935, 802798, 805237, 812669, 817212, 817904, 819548, 822249, 830291, 838259, 846837, 849237, 851349, 860556, 865248, 874512, 891826, 903335, 919083, 922789, 923219, 924617, 931960, 941218, 946285, 968134, 975820, 984051, 988826, 994495, 996189, 1000966, 1008789, 1017419, 1026127, 1029167, 1055772, 1057771, 1072702, 1084128, 1089763, 1113852, 1128875, 1134903, 1152880, 1153151, 1161083, 1162743.

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Code availability. The C++ sieve, Python pipeline scripts, and all raw data files used in this paper are publicly available at `https://github.com/GUTgeoservice/Q5-primes`.

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